CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 13: Factorisation

Every formula and result of the chapter on one sheet

1	Common factor	$ab + ac = a(b + c)$
2	Grouping	$ax + ay + bx + by = (a + b)(x + y)$
3	Difference of squares	$a^2 - b^2 = (a + b)(a - b)$
4	Perfect square	$a^2 + 2ab + b^2 = (a + b)^2$
5	Perfect square	$a^2 - 2ab + b^2 = (a - b)^2$
6	Monic trinomial	$x^2 + (p + q)x + pq = (x + p)(x + q)$
7	General trinomial	Split bx using numbers with product ac and sum b
8	Sign change	$b - a = -(a - b)$
9	Fourth powers	$a^4 - b^4 = (a^2 + b^2)(a + b)(a - b)$
10	Division	Factorise and cancel common factors
11	Check	Multiply the factors to get the original expression